A Siamese network for identifying gravitationally lensed repeating fast radio bursts: benchmark, transit geometry and forecasts for CHIME/FRB Catalog 2 and all-sky monitors

Draft authors *
Affiliations to be added

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ABSTRACT

A galaxy-lensed repeating fast radio burst (FRB) would repeat every burst after a fixed delay of days to months, and the two copies of a burst would differ only by propagation effects. We build semi-synthetic lensed repeaters from the 981 public dynamic spectra of the 83 repeaters in CHIME/FRB Catalog 2, with real off-pulse noise, measured injection S/N and per-image scattering, scintillation and residual dispersion, plus *impostor* pairs of different bursts that share a plasma-lens envelope or a scintillation pattern. On this benchmark a Siamese encoder with a correlation layer and feature-map fusion recovers 0.47 [0.25, 0.66] of lensed pairs at 1% false-positive rate in source-grouped cross-validation, against 0.25 [0.20, 0.30] for the best physics- or feature-based baseline (paired difference +0.22 [+0.04, +0.38]; development data, test split unopened). Its residual false positives are almost all envelope-sharing impostors, and ablations show that both the alignment layer and training on impostors are needed. On real cross-source pairs, which are true negatives, its false-positive rate at a threshold set on per-pair simulated twins is 1.0–1.5% over three seeds, within the pre-registered 2% limit, and two copies of a real burst are recognised as often as two copies of its smooth template. An independent reproduction regenerates all frozen sets bit for bit. At the source level, the delay of a lens is constant to far better than the 4 ms catalogue timing, so two burst pairs sharing one delay are already significant for all but the most active repeaters. However, a transit telescope sees a burst only within minutes of transit, so the copy of every recorded burst lands at the same hour angle: copies are recorded only if the delay is within minutes of a whole number of sidereal days. This happens for 1.8% of lens delays at the median declination of the sample, so lensed repeaters do *not* escape the transit problem that limits one-off FRBs. For Catalog 2 we expect 1.2×10^{-4} detectable lensed repeaters (5.3×10^{-6} without magnification bias), and an all-sky monitor watching the same sources would gain a factor ∼50; a longer transit survey would not.

Key words: fast radio bursts – gravitational lensing: strong – methods: data analysis – methods: statistical

1 INTRODUCTION

Strongly lensed FRBs have been proposed as probes of compact dark matter (Kader et al. 2022; Leung et al. 2022; Krochek & Kovetz 2022) and, for galaxy lenses with delays of days to months, of cosmology through precise time delays (Li et al. 2018; Wucknitz et al. 2021). No lensed FRB is confirmed; the strongest candidate is a 3.4σ millisecond-scale double burst in Catalog 1 (Chang et al. 2024). For one-off FRBs a transit instrument such as CHIME rarely records both images of a galaxy lens: the second image must arrive while the source is again in the beam (Connor & Ravi 2023). Repeating FRBs have been argued to avoid this, because both image streams keep firing over years of exposure.

Two questions decide whether a search of the CHIME/FRB Catalog 2 repeaters is useful. First, can the morphology of two bursts tell a lensed copy from a different burst of the same source, including impostors that share a propagation imprint? Second, how often are both copies of a burst recorded at all? Machine-learning work on FRBs has so far classified repeaters against non-repeaters; lensed-pair classifiers exist for gravitational waves (Goyal et al. 2021; Zhang et al. 2026) but not for FRBs. We answer both questions with real CHIME data wherever it exists and simulation only for the lensing itself. Our contributions are: (i) a semi-synthetic benchmark of lensed, repeat, unrelated and impostor burst pairs built from all 981 public Catalog 2 repeater spectra, with a ladder of physics, feature and learned methods; (ii) a transfer check on real true-negative pairs in which per-pair twin re-simulation separates genuine sim-to-real gaps from sample composition; (iii) a source-level timing test with a held-out false-alarm calibration; (iv) the transit-alignment argument, verified by simulation against the analytic window width, which shows that lensed repeaters do not escape the transit problem; and (v) forecasts for Catalog 2, a longer survey and an all-sky monitor. All development results were reproduced on a second ma-

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chine, and the test split remains sealed under a pre-registered blind protocol.

2 DATA

We use the Stokes-I dynamic spectra of all 981 repeater bursts (83 sources) in CHIME/FRB Catalog 2 (CHIME/FRB Collaboration 2026), at 0.983 ms and 400–800 MHz. Each burst is binned to 256 channels × 160 samples, centred on the catalogue emission region, and normalised by its off-pulse statistics; six bursts at coarser time resolution are excluded as templates. The fitburst models of the bursts serve as noise-free templates. They miss a median 35% of the on-pulse signal energy, which lowers the similarity of simulated pairs of every kind by a near-constant ∼8% relative to real pairs in a twin test, while for the cross-correlation baseline the simulated rank correlation with real pairs reaches the simulation-to-simulation noise ceiling in every S/N bin (the learned model does not; Section 5.1). Absolute thresholds therefore come from real negatives only. Sources are split 60/20/20 into train/validation/test by burst count, with the two most active repeaters forced into training. An independent reproduction on a second machine, which re-downloaded and reprocessed all 981 spectra, rebuilt the injection cache with the same counts (950 eligible templates, 856 backgrounds) and regenerated the frozen validation and grouped-fold sets: hashes 7/7 match.

3 SEMI-SYNTHETIC LENSED PAIRS

A pair is one of four classes. A *lensed* pair places the same template twice; a *repeat* pair places two different templates of one source; an *unrelated* pair takes templates from two sources; an *impostor* pair places two different templates of one source that share either a smoothed spectral envelope (I1, a plasma-lens caustic), a scintillation pattern (I2), or both. Each member is injected at a measured matched-filter S/N on a real off-pulse background from another burst, with its own time shift (±10 samples), residual DM, scattering and scintillation; flux ratios are drawn so that they carry no information. Frozen validation and test sets of 4000 pairs and bright sets with pair minimum S/N ≥ 25 (the pre-specified headline bin) are hashed so that any change of code or inputs is detected.

4 METHODS

4.1 Benchmark ladder

All methods score the same pairs, averaged over both member orders: (1) cross-correlation within the on-pulse box, maximised over lags and common channels; (2) a physics statistic fitting one member to the other convolved with a scattering kernel over μ, lag, ΔDM and scattering grids; (3) gradient-boosted trees on seven pixel-measured burst properties; (4) a two-channel CNN; (5) a classic Siamese network (vector embeddings); (6) a Siamese encoder whose feature maps are aligned by a correlation layer over ±24 samples and fused as $|g(A) - g(B)|$ and $g(A) \cdot g(B)$ (primary); (7) cross-attention between time–frequency patches. Learned models are trained on pairs generated on the fly from training sources with a four-class cross-entropy.

4.2 Metric and comparison

The primary metric is recall of lensed pairs at 1% false-positive rate against repeat and impostor pairs, with a source-level bootstrap in which thresholds are recomputed in every resample. Because the validation CI was wide, the pre-registered rule switched the primary comparison to five source-grouped folds before the test split is opened.

4.3 Source-level timing (Track C)

A lens repeats every burst after one delay Δt. Catalogue arrival times referenced to infinite frequency have a median uncertainty of 4 ms, so pairs (i, j) and (k, l) of one source share a delay if $|\Delta t_{ij} - \Delta t_{kl}| < 4\sqrt{\sigma_{ij}^2 + \sigma_{kl}^2}$ and their S/N ratios agree within a factor 2. The statistic $k_{\max}$ is the largest set of disjoint pairs sharing a delay. Its background comes from time-scrambled burst lists, in which every burst moves by a random whole number of sidereal days; this keeps hour angles, S/N and the transit selection. The unscrambled lists are never scored before unblinding. Each source's threshold $k_{\rm thr}$ is the smallest $k \geq 2$ whose background probability is at most 10^{-3}, set from 2000 scrambles. Efficiencies come from lensed copies injected into scrambled lists.

4.4 Lens population and observation

Galaxy lenses are singular isothermal spheres with the velocity-dispersion function of Choi et al. (2007). Source redshifts come from host galaxies for seven localised repeaters and otherwise from the NE2001 DM excess through the Macquart relation (Macquart et al. 2020), with halo and host contributions. For each lens, $\Delta t = 2y\theta_{\rm E}^2(1 + z_{\rm l})D_{\rm l}D_{\rm s}/(cD_{\rm ls})$ and the trailing image is fainter by $r = (1 - y)/(1 + y)$. A copy is recorded if it falls within $|{\rm HA}| \cos\delta < 1.1^\circ$ of upper transit (or lower transit for $\delta > 70^\circ$), inside the survey, on an observing day (from the catalogue exposure), and above S/N 10. The transit window was measured from the real bursts: 99% of them lie within 1.08° of transit.

5 RESULTS

5.1 Pair classification

Table 1 and Fig. 1 summarise the development results. Model 6 leads every other method on the fixed validation split and in grouped folds. Replicates on a second machine reach 0.668 / 0.649 / 0.616 (Table 1), about 0.05 below the original seeds but still far above the GBM. Its false positives at the 1% threshold are almost all I1 impostors (shared spectral envelope); I2 impostors and plain repeats are rarely confused. One two-burst source sets the threshold of one fold, because its few repeat and impostor pairs look alike to the model, which widens the out-of-fold interval.

Table 1. Recall at 1% FPR, lensed vs repeat + impostor, S/N $\geq$ 25, development data. Brackets: 95% source bootstrap.

Method	Fixed split (3 seeds)	Grouped folds (OOF)
Feature GBM (best of 1–3)	0.285 [0.180, 0.401]	0.246 [0.199, 0.297]
Model 6 (correlation + fusion)	0.701 / 0.698 / 0.684	0.467 [0.253, 0.662]
Model 5 (classic Siamese)	0.500 / 0.559 / 0.541	–
Model 4 (two-channel CNN)	0.423 / 0.409 / 0.407	–
Model 7 (cross-attention)	0.471 / 0.013 / 0.579	–
Model 6 − GBM	+0.394 [+0.226, +0.542]	+0.220 [+0.035, +0.381]
Model 6, independent replicate[a]	0.668 / 0.649 / 0.616	–

[a]Second machine, seeds 0/1/2, 4 data workers instead of 14, which changes the training pair stream. The pre-registered range 0.66–0.72 applied to seed 0 (passed); seeds 1 and 2 fall below it, so this machine's replicates sit about 0.05 below the original seeds.

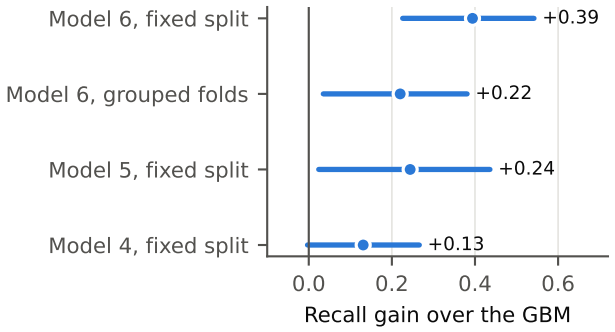


Figure 1. Paired recall gain over the feature GBM at 1% FPR (S/N $\geq$ 25), with 95% source-bootstrap intervals (development data; fixed split: three seeds; grouped folds: out-of-fold).

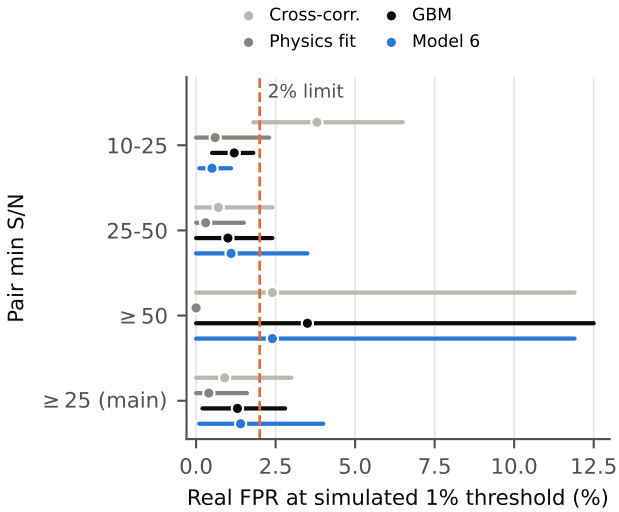


Figure 2. Stage 3 transfer check on validation sources: share of real cross-source pairs (true negatives) above each method's threshold at 1% FPR on simulated unrelated pairs (model 6: seed 0; blocks of size 4 for the baselines), by pair minimum S/N ('main': the pre-registered S/N $\geq$ 25 bin), with 95% source-bootstrap intervals. Dashed: the pre-registered 2% limit.

Transfer to real data. Cross-source pairs of real bursts are true negatives, so they can be scored without unblinding. Among the 89 validation bursts of 26 sources there are 3750 such pairs. For each method and bin of pair minimum S/N and mask overlap we set the threshold at 1% FPR on 20 000 generic simulated unrelated pairs from the same sources, and measured the real rate above it (pre-registered criterion: at most 2%). In the headline bin (793 real pairs) every method passes (Fig. 2): the real FPR is 1.4% [0.1, 4.0] for model 6 (1.4, 1.9, 1.6% for seeds 0, 1, 2; intervals reach 5%) and 1.3% [0.2, 2.8] for the GBM (95% source bootstrap). Two cells fail. Cross-correlation exceeds 2% below S/N 25 (6.2, 3.8, 2.4% for the three block sizes), and in the S/N $\geq$ 50 bin (85 real pairs) several methods exceed it in point estimate through the same two pairs, both involving the bright bursts of FRB 20220428C.

Transfer versus composition. A real-negative rate can differ from 1% for two reasons: the simulator misses something (a transfer gap), or the real pairs are a different mix of templates and S/N from the generic simulated pairs (composition). The twins separate the two: each real pair is re-simulated twice from its own two templates at its own S/N, so at the same threshold the real-minus-twin rate is the transfer gap and the twin rate minus 1% is the composition effect. For every baseline the transfer gap is consistent with zero in every bin (at most 0.5 points in the headline bin). The cross-correlation failure below S/N 25 is pure composition: its twins exceed the threshold as often as the real pairs (5.6, 4.0, 2.5%), because real pairs in that bin are brighter (median S/N 16.9 against 13.5) and the cross-correlation grows with S/N within the bin. With the team's twin-matched threshold and real windows whose zero-padded columns are filled like the simulated backgrounds, every method passes in both bins on this machine as on the original one: model 6 reaches 1.5, 1.3 and 1.0% in the headline bin for three seeds, and its rank correlation with its twins (0.91–0.92) reaches the twin-to-twin ceiling (0.87–0.89). An earlier version of this check, without the fill, attributed a small model-6 excess to template incompleteness; that excess came from the zero-padded windows and is withdrawn. Thresholds for a search should therefore be twin-matched and computed on filled real windows.

Ablations. Each ablation retrains model 6 on this machine and is compared with the full model trained with the same seeds (3 per configuration; full model 0.644), as seed-mean recall on the validation bright set with paired source-bootstrap intervals. Removing the correlation layer, so that feature maps are fused without alignment, gives recall 0.557, dif-

Table 2. Expected lensed repeaters in the Catalog 2 sample. γ: magnification-bias slope of the integral counts.

Survey	$\sum\tau$	$N_{\rm exp}$ (timing)	$N_{\rm exp}$ (≥ 1 copy)
CHIME, Catalog 2	2.9×10^{-3}	5.3×10^{-6}	3.2×10^{-5}
All-sky, same sources	2.9×10^{-3}	2.6×10^{-4}	6.3×10^{-4}
CHIME, Catalog 2, $\gamma=1.5$	1.8×10^{-2}	1.2×10^{-4}	4.9×10^{-4}
All-sky, same sources, $\gamma=1.5$	1.8×10^{-2}	6.1×10^{-3}	9.3×10^{-3}
CHIME-like, 10 yr	2.8×10^{-3}	7.4×10^{-6}	4.9×10^{-5}
All-sky, 10 yr	2.8×10^{-3}	3.7×10^{-4}	6.8×10^{-4}
CHIME-like, 10 yr, $\gamma=1.5$	1.8×10^{-2}	2.2×10^{-4}	6.3×10^{-4}
All-sky, 10 yr, $\gamma=1.5$	1.8×10^{-2}	7.4×10^{-3}	9.6×10^{-3}

ference -0.087 [-0.186, -0.025]. Removing impostors from training gives 0.474, difference -0.170 [-0.254, -0.052]: without them the network mistakes envelope-sharing bursts of one source for lensed copies. Oversampling hard repeat negatives (the most template-similar of four draws) gives 0.698, difference $+0.053$ [-0.011, $+0.082$]: a small gain in every seed whose interval still includes zero. Hard-negative training also biases a learned likelihood ratio (Appendix A), so we do not adopt it without a correction. The two design choices that distinguish model 6, alignment and an impostor class, are thus both needed.

5.2 Timing false-alarm rate

To check the thresholds, we recomputed every $k_{\rm thr}$ from one random stream and counted exceedances in 2000 fresh scrambles per source from an independent stream. We found 1 of 166,000 held-out scrambles (95% upper limit 2.9×10^{-5} per source and list), or at most 0.002 false sources in a search of all 83 repeaters. The test is conservative because k is an integer: for most sources no scramble reaches even $k=2$. Only the most active repeater, FRB 20220912A (373 bursts), reaches $k=2$ in every scramble and needs $k_{\rm thr}=4$.

5.3 Transit geometry

Every burst that CHIME records arrives within minutes of transit. Its copy arrives Δt later, at hour angle HA + 24 h $\times$ frac($\Delta t/T_{\rm sid}$), the *same* offset for every burst of the source. Copies are therefore recorded for all bursts or for none, and only if Δt lies within a few minutes of a whole number of sidereal days. Over the SIS delay distribution this happens for a fraction $p_{\rm align}$ of lenses (median 1.8% over the 83 sources; higher near the pole, where the window widens and lower transits add a chance near half a sidereal day). A repeater thus offers many pairs *once aligned*, but the chance of alignment is the same $\sim\theta_{\rm EW}/2\pi$ that limits one-off FRBs (Connor & Ravi 2023). This corrects the assumption that lensed repeaters escape the transit problem. Fig. 3 shows that the simulated $p_{\rm align}$ follows the width of two transit windows per sidereal day, the analytic expectation when the recorded bursts spread across one window.

5.4 Forecasts

Table 2 gives the expected number of lensed repeaters whose timing statistic exceeds a per-source false-alarm probability of 10^{-3}, and the number with at least one recorded copy (a candidate for the pair classifier). Fig. 4 compares CHIME with an all-sky monitor watching the same sources. A null result in Catalog 2 is therefore expected, and the search

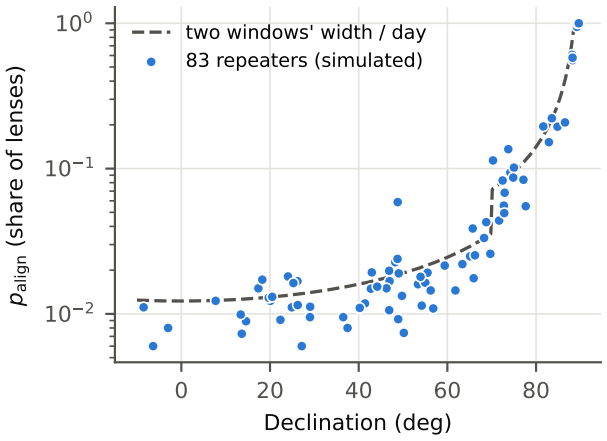


Figure 3. Share of SIS lens delays for which CHIME records the copies of a repeater's bursts, against declination. Points: the 83 Catalog 2 repeaters with their own burst hour angles, exposure and redshift. Dashed: twice the transit-window width per sidereal day (doubled again above 70°, where lower transits add a window).

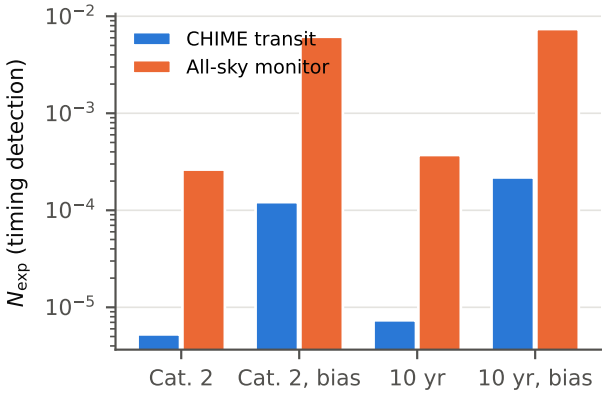


Figure 4. Expected timing detections in the Catalog 2 repeater sample, for the survey as observed and stretched to 10 years, without and with magnification bias ($\gamma=1.5$).

serves as a demonstration and a check of the false-alarm rate. Stretching the same transit survey to 10 years raises $N_{\rm exp}$ by only a factor 1.4–1.8: a longer baseline admits longer delays and more bursts, but leaves the alignment requirement unchanged. An all-sky monitor watching the same sources gains a factor $\sim$50. For CHIME-like surveys the limit is sky coverage, not time on sky.

6 DISCUSSION

What the classifier can and cannot claim. Simulated false-positive rates rank methods only; thresholds for a search come from real negatives (cross-source pairs and time-scrambled lists). The lensed class has no real counterpart, so its realism rests on the twin tests and the template-deficit ar-

gument, which makes simulated lensed pairs *less* alike than real ones would be.

From scores to evidence. Posterior odds for a candidate are a likelihood ratio times prior odds from τ. On a closed-form benchmark (Appendix A) we find that an uncorrected hard-negative training scheme biases a learned likelihood ratio low by about 1 dex, that a misspecified repeat-similarity prior raises the false-positive rate about 12-fold at a fixed threshold, and that the identity $E_{H_0}[r] \leq 1$ can be tested on known negatives alone with a betting e-process (Waudby-Smith & Ramdas 2024; Ramdas & Wang 2024). We applied the identity to model 6, taking $\hat{r} = \exp(\ell_{\rm lensed} - \ell_{\rm unrelated})$ from its balanced-class logits. On simulated unrelated pairs the mean of $\hat{r}$ is 1.9, 1.7, 7.3 for seeds 0, 1, 2 (seed 0: bootstrap 1.42–2.61), so the network over-claims evidence by a factor of about two or more, driven by a handful of pairs in the tail. On the 3750 real cross-source pairs the means are 1.10, 1.86, 1.53 (seed 0: 1.10 [0.20, 2.55], source bootstrap), consistent with 1 within a factor of two: real negatives show no extra over-claiming. The betting test has little power here, because $\hat{r}$ is heavy-tailed (the ten largest values carry about half the sum), so we report the bootstrap. A factor-of-a-few bias in $\hat{r}$ is small against the uncertainty of τ, but a posterior for any candidate should calibrate $\hat{r}$ on real negatives first.

Implications for future surveys. An all-sky monitor removes the alignment requirement, and a longer transit survey does not (Section 5.4). Arrays with wide, continuous fields of view, such as the proposed BURSTT (Ho et al. 2023), are therefore the natural place for repeater-timing searches. A rate per year for such an array needs its own repeater population and exposure and is beyond this paper; the per-source efficiencies we publish allow it to be computed. For transit instruments the pair classifier keeps a role that timing cannot fill: a single recorded copy of a single burst, which can occur even when no timing pattern forms. In the Catalog 2 sample we expect 3.2×10^{-5} lensed repeaters with at least one recorded copy, about six times the timing yield; at the development recall of model 6 the pair channel would find 1.5×10^{-5}. Both channels are far below one for CHIME, so the classifier's value lies in surveys with continuous sky coverage and in vetting candidates, where its false positives (envelope-sharing impostors) are the relevant background.

Limitations. (i) All classification numbers are development results; the test split stays sealed until the team signs off the propagation model. (ii) Lenses are singular isothermal spheres without shear, so four-image systems and shear-induced delay spreads are absent; the transit argument depends only on the delay, not on the lens model. (iii) Redshifts of the 76 unlocalised repeaters come from DM, and a few high-DM sources dominate $\sum\tau$; a large host DM would bias them high. Raising the median host DM from 68 to 150 pc cm^{-3} changes $\sum\tau$ by -13% and the CHIME timing yield by -19%, so the forecasts are not sensitive to it. (iv) The magnification-bias slope of CHIME repeaters is unmeasured, which is why we bracket it. (v) Observing days are drawn from the integrated exposure, assuming downtime spread evenly over the survey. (vi) The Stage 3 check uses 26 validation sources, so its S/N ≥ 50 bin is small.

7 CONCLUSIONS

(i) On semi-synthetic pairs built from all public Catalog 2 repeater spectra, a Siamese encoder with a correlation layer and feature-map fusion separates lensed copies from other bursts of the same source better than physics and feature baselines: recall 0.47 [0.25, 0.66] against 0.25 [0.20, 0.30] at 1% FPR in source-grouped folds (development data). Envelope-sharing impostors are the remaining confusion. On real cross-source negatives its false-positive rate at a twin-matched threshold is 1.0–1.5% over three seeds, within the pre-registered limit, once zero-padded real windows are filled; twin matching also explains the one baseline failure under generic thresholds as sample composition.

(ii) A galaxy lens fixes the delay to far better than the catalogue timing, so two burst pairs sharing one delay are already a detection for all but the most active repeaters. Held-out scrambles confirm the false-alarm rate is conservative: at most 0.002 false sources in a search of all 83 repeaters.

(iii) A transit instrument records the copies of a repeater's bursts only if the delay is within minutes of a whole number of sidereal days, which happens for 1.8% of lenses at the median. Lensed repeaters therefore do not escape the transit problem of one-off FRBs.

(iv) The Catalog 2 sample is expected to contain 5.3×10^{-6}–1.2×10^{-4} detectable lensed repeaters, so a null search is expected. A longer transit survey changes this little; an all-sky monitor gains a factor $\sim$50.

[Test-split and search results, after team sign-off under the blind protocol.]

DATA AVAILABILITY

All inputs are public (CHIME/FRB Catalog 2, DOI 10.11570/25.0066). Code, frozen manifests and hashes are in the project repository.

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APPENDIX A: REFERENCE LIKELIHOOD RATIOS ON A CLOSED-FORM BENCHMARK

A linear-Gaussian pair simulator with an exact marginal likelihood ratio (lensed: correlation $\rho = 1$ between the intrinsic shapes; repeat: $\rho < 1$ drawn from a prior) tests neural ratio estimators against the truth on 2.4 million pairs. Standard training is self-calibrated in the tail and loses power instead; doubling the training budget recovers much of it. Oversampling similar negatives without correction under-claims by about 1 dex at every budget, while importance weighting or conditioning on ρ and re-marginalising preserve calibration. Full results: update 2026-11-07 in the repository.

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